

Configuración Entorno virtualizado con Vagrant

Jesús David Morales Muriel

Para este trabajo se logran adjuntar las evidencias de los pasos tomados, así mismo se adjunta los links de la virtual box y el repositorio.

```
vagrant@servidor:~$ sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
vagrant@servidor:~$ sudo apt-get install -y git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn
The following packages will be upgraded:
  git
1 upgraded, 0 newly installed, 0 to remove and 157 not upgraded.
Need to get 3,174 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git amd64 1:2.34.1-1ubuntu1.17 [3,174 kB]
Fetched 3,174 kB in 1s (2,732 kB/s)
(Reading database ... 45872 files and directories currently installed.)
Preparing to unpack .../git_1%3a2.34.1-1ubuntu1.17_amd64.deb ...
Unpacking git (1:2.34.1-1ubuntu1.17) over (1:2.34.1-1ubuntu1.15) ...
Setting up git (1:2.34.1-1ubuntu1.17) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

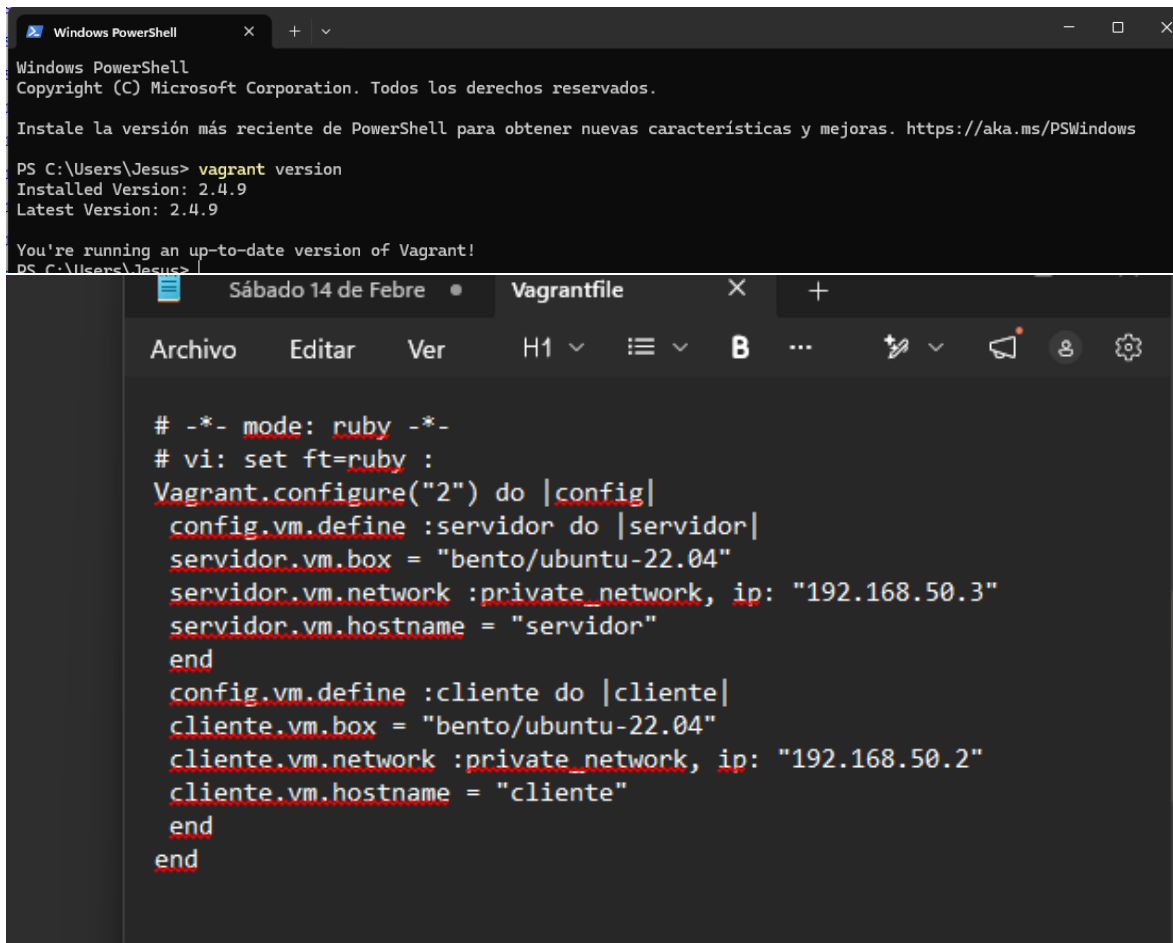
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
vagrant@servidor:~$ git config --global user.name "JesusMorales-DS"
vagrant@servidor:~$ git clone https://github.com/JesusMorales-DS/Telematica
Cloning into 'Telematica'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
vagrant@servidor:~$ cd Telematica
vagrant@servidor:~/Telematica$ |

vagrant@servidor:~/Telematica$ git push origin Practicas
Username for 'https://github.com': JesusMorales-DS
Password for 'https://JesusMorales-DS@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 461 bytes | 461.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/JesusMorales-DS/Telematica
   e2faa28..c522ee3  Practicas -> Practicas
vagrant@servidor:~/Telematica$ |
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\Jesus> vagrant version
Installed Version: 2.4.9
Latest Version: 2.4.9

You're running an up-to-date version of Vagrant!
PS C:\Users\Jesus>
```

```
# -*- mode: ruby -*-
# vi: set ft=ruby :
Vagrant.configure("2") do |config|
  config.vm.define :servidor do |servidor|
    servidor.vm.box = "bento/ubuntu-22.04"
    servidor.vm.network :private_network, ip: "192.168.50.3"
    servidor.vm.hostname = "servidor"
  end
  config.vm.define :cliente do |cliente|
    cliente.vm.box = "bento/ubuntu-22.04"
    cliente.vm.network :private_network, ip: "192.168.50.2"
    cliente.vm.hostname = "cliente"
  end
end
```

```
PS C:\Users\Jesus> cd prueba
PS C:\Users\Jesus\prueba> vagrant init
A 'Vagrantfile' has been placed in this directory. You are now
ready to 'vagrant up' your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
'vagrantup.com' for more information on using Vagrant.
PS C:\Users\Jesus\prueba> vagrant up
Bringing machine 'servidor' up with 'virtualbox' provider...
Bringing machine 'cliente' up with 'virtualbox' provider...
==> servidor: Box 'bento/ubuntu-22.04' could not be found. Attempting to find and install...
servidor: Box Provider: virtualbox
servidor: Box Version: >= 0
==> servidor: Loading metadata for box 'bento/ubuntu-22.04'
servidor: URL: https://vagrantcloud.com/api/v2/vagrant/bento/ubuntu-22.04
==> servidor: Adding box 'bento/ubuntu-22.04' (v202510.26.0) for provider: virtualbox (amd64)
servidor: Downloading: https://vagrantcloud.com/bento/boxes/ubuntu-22.04/versions/202510.26.0/providers/virtualbox/amd64/vagrant.box
servidor:
==> servidor: Successfully added box 'bento/ubuntu-22.04' (v202510.26.0) for 'virtualbox (amd64)'!
==> servidor: Importing base box 'bento/ubuntu-22.04'...
==> servidor: Matching MAC address for NAT networking...
==> servidor: Checking if box 'bento/ubuntu-22.04' version '202510.26.0' is up to date...
==> servidor: Setting the name of the VM: prueba_servidor_1784665696231_6274
Vagrant is currently configured to create VirtualBox synced folders with
the 'SharedFoldersEnableSymlinksCreate' option enabled. If the Vagrant
guest is not trusted, you may want to disable this option. For more
information on this option, please refer to the VirtualBox manual:

https://www.virtualbox.org/manual/ch04.html#sharedfolders

This option can be disabled globally with an environment variable:

VAGRANT_DISABLE_VBOXSYMLINKCREATE=1

or on a per folder basis within the Vagrantfile:

config.vm.synced_folder '/host/path', '/guest/path', SharedFoldersEnableSymlinksCreate: false
==> servidor: Clearing any previously set network interfaces...
==> servidor: Preparing network interfaces based on configuration...
servidor: Adapter 1: nat
servidor: Adapter 2: hostonly
==> servidor: Forwarding ports...
servidor: 22 (guest) => 2222 (host) (adapter 1)
==> servidor: Booting VM...
==> servidor: Waiting for machine to boot. This may take a few minutes...
servidor: SSH address: 127.0.0.1:2222
servidor: SSH username: vagrant
```

Current machine states:

servidor	running (virtualbox)
cliente	running (virtualbox)

This environment represents multiple VMs. The VMs are all listed above with their current state. For more information about a specific VM, run 'vagrant status NAME'.

```
root@cliente:~# ping 192.168.50.3
PING 192.168.50.3 (192.168.50.3) 56(84) bytes of data.
64 bytes from 192.168.50.3: icmp_seq=1 ttl=64 time=1.10 ms
64 bytes from 192.168.50.3: icmp_seq=2 ttl=64 time=0.698 ms
64 bytes from 192.168.50.3: icmp_seq=3 ttl=64 time=0.766 ms
64 bytes from 192.168.50.3: icmp_seq=4 ttl=64 time=2.50 ms
64 bytes from 192.168.50.3: icmp_seq=5 ttl=64 time=0.639 ms
64 bytes from 192.168.50.3: icmp_seq=6 ttl=64 time=0.588 ms
64 bytes from 192.168.50.3: icmp_seq=7 ttl=64 time=0.654 ms
64 bytes from 192.168.50.3: icmp_seq=8 ttl=64 time=0.709 ms
64 bytes from 192.168.50.3: icmp_seq=9 ttl=64 time=0.645 ms
64 bytes from 192.168.50.3: icmp_seq=10 ttl=64 time=1.03 ms
64 bytes from 192.168.50.3: icmp_seq=11 ttl=64 time=0.609 ms
64 bytes from 192.168.50.3: icmp_seq=12 ttl=64 time=0.566 ms
64 bytes from 192.168.50.3: icmp_seq=13 ttl=64 time=0.560 ms
64 bytes from 192.168.50.3: icmp_seq=14 ttl=64 time=0.691 ms
64 bytes from 192.168.50.3: icmp_seq=15 ttl=64 time=0.563 ms
64 bytes from 192.168.50.3: icmp_seq=16 ttl=64 time=0.689 ms
64 bytes from 192.168.50.3: icmp_seq=17 ttl=64 time=0.668 ms
64 bytes from 192.168.50.3: icmp_seq=18 ttl=64 time=0.611 ms
64 bytes from 192.168.50.3: icmp_seq=19 ttl=64 time=0.555 ms
64 bytes from 192.168.50.3: icmp_seq=20 ttl=64 time=0.620 ms
64 bytes from 192.168.50.3: icmp_seq=21 ttl=64 time=1.05 ms
64 bytes from 192.168.50.3: icmp_seq=22 ttl=64 time=0.737 ms
64 bytes from 192.168.50.3: icmp_seq=23 ttl=64 time=0.697 ms
64 bytes from 192.168.50.3: icmp_seq=24 ttl=64 time=0.661 ms
64 bytes from 192.168.50.3: icmp_seq=25 ttl=64 time=0.664 ms
64 bytes from 192.168.50.3: icmp_seq=26 ttl=64 time=0.581 ms
64 bytes from 192.168.50.3: icmp_seq=27 ttl=64 time=0.970 ms
64 bytes from 192.168.50.3: icmp_seq=28 ttl=64 time=1.16 ms
64 bytes from 192.168.50.3: icmp_seq=29 ttl=64 time=0.719 ms
64 bytes from 192.168.50.3: icmp_seq=30 ttl=64 time=0.695 ms
64 bytes from 192.168.50.3: icmp_seq=31 ttl=64 time=0.578 ms
64 bytes from 192.168.50.3: icmp_seq=32 ttl=64 time=0.677 ms
64 bytes from 192.168.50.3: icmp_seq=33 ttl=64 time=0.653 ms
64 bytes from 192.168.50.3: icmp_seq=34 ttl=64 time=0.616 ms
64 bytes from 192.168.50.3: icmp_seq=35 ttl=64 time=0.495 ms
64 bytes from 192.168.50.3: icmp_seq=36 ttl=64 time=1.46 ms
64 bytes from 192.168.50.3: icmp_seq=37 ttl=64 time=0.657 ms
64 bytes from 192.168.50.3: icmp_seq=38 ttl=64 time=0.689 ms
64 bytes from 192.168.50.3: icmp_seq=39 ttl=64 time=0.633 ms
64 bytes from 192.168.50.3: icmp_seq=40 ttl=64 time=0.538 ms
|
```

```

PS C:\Users\Jesus\prueba> vagrant ssh servidor
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-160-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jul 21 11:59:11 PM UTC 2026

System load:            0.26
Usage of /:              16.2% of 30.34GB
Memory usage:           8%
Swap usage:              0%
Processes:              159
Users logged in:        0
IPv4 address for eth0: 10.0.2.15
IPv6 address for eth0: fd17:625c:f037:2:a00:27ff:fe16:ddef

This system is built by the Bento project by Chef Software
More information can be found at https://github.com/chef/bento

Use of this system is acceptance of the OS vendor EULA and License Agreements.
Last login: Tue Jul 21 20:47:48 2026 from 10.0.2.2
vagrant@servidor:~$ cd /vagrant
vagrant@servidor:/vagrant$ ls -l
total 1017872
-rwxrwxrwx 1 vagrant vagrant 1042292968 Jul 21 21:03 mynew.box
-rwxrwxrwx 1 vagrant vagrant      0 Jul 21 22:04 prueba_sincro.txt
-rwxrwxrwx 1 vagrant vagrant    448 Jul 21 20:25 Vagrantfile
vagrant@servidor:/vagrant$ cat prueba_sincro.txt
vagrant@servidor:/vagrant$ echo "hola desde la maquina Virtual" > creado_en_vm.txt
vagrant@servidor:/vagrant$ exit
logout
PS C:\Users\Jesus\prueba> ls

Directorio: C:\Users\Jesus\prueba

Mode                LastWriteTime         Length Name
----                -
d-----          21/07/2026   3:25 p. m.         .vagrant
-a-----          21/07/2026   7:01 p. m.           30 creado_en_vm.txt
-a-----          21/07/2026   4:03 p. m.    1042292968 mynew.box
-a-----          21/07/2026   5:04 p. m.           0 prueba_sincro.txt
-a-----          21/07/2026   3:25 p. m.       448 Vagrantfile

PS C:\Users\Jesus\prueba> |

```

```

root@servidor:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe16:ddef prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe16:ddef prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:16:dd:ef txqueuelen 1000 (Ethernet)
    RX packets 63382 bytes 92767618 (92.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10333 bytes 731154 (731.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.50.3 netmask 255.255.255.0 broadcast 192.168.50.255
    inet6 fe80::a00:27ff:fee6:21ac prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:e6:21:ac txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 243 (243.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 1096 (1.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 66 bytes 8603 (8.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 66 bytes 8603 (8.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@cliente:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe16:ddef prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe16:ddef prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:16:dd:ef txqueuelen 1000 (Ethernet)
    RX packets 26822 bytes 37802621 (37.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 6634 bytes 531641 (531.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.50.2 netmask 255.255.255.0 broadcast 192.168.50.255
    inet6 fe80::a00:27ff:fe8e:74a3 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8e:74:a3 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13 bytes 1006 (1.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 42 bytes 5759 (5.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 42 bytes 5759 (5.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

vagrant@servidor:~$ cd Telematica
vagrant@servidor:~/Telematica$ mkdir -p mispracticas/Practica0 mispracticas/Practical mispracticas/Practica2
vagrant@servidor:~/Telematica$ echo "Reporte de la Practica 1 - Entorno Virtualizado con Vagrant" > mispracticas/Practical/README.md
vagrant@servidor:~/Telematica$ git add .
vagrant@servidor:~/Telematica$ git commit -m "Estructura inicial de carpetas de practicas"
[Practicass c522ee3] Estructura inicial de carpetas de practicas
Committer: JesusMorales-DS <vagrant@servidor>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

--- 192.168.50.3 ping statistics ---
105 packets transmitted, 105 received, 0% packet loss, time 105928ms
rtt min/avg/max/mdev = 0.423/0.712/3.123/0.346 ms
root@cliente:~# exit
logout

```